COLOUR AND COMPUTERS

AN OVERVIEW

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Through cameras, scanners, screens and printers, what you see is not always what you get. Here we introduce the theory underlying colour management

Colour is something that most of us take for granted, but for manufacturers, lack of colour control can be a serious economic problem. Visible inconsistency in paints, inks and dyes costs money. For example, a car manufacturer couldn't sell cars if a particular paint colour varied from one car to the next. For product advertising, manufacturers need be able accurately to represent a product's colour in print, on the Internet, in film and on television. For this reason, as soon as it was possible for industries to produce a wide range of colour pigments, our perception of colour, and the measurement and control of colour, became the subject of a great deal of study.

At home and at work even computer users who aren't colour professionals are finding that they need to manipulate colour images. Intentional colour management provides the means both to control and reproduce colours as accurately as possible.

To apply colour management effectively you need to know the colour theory behind it. This article introduces some of the ideas of colour theory and shows how it is used in digital colour management.

A Brief History Of Colour

Humans have been using coloured pigments for thousands of years, although for much of this time only a limited range of colours were available. The use of oils blended with new pigments by Jan van Eyck, best known for his 1434 painting of "Giovanni Arnolfini and his wife", revolutionised painting. However it is only since the eighteenth century that a really wide range of low cost colour pigments have been available. The Industrial Revolution saw the invention of aniline dies from coal products in the 1800s and the first colour photograph was made in 1861.

The earliest colour space, still in use, the CIE XYZ space, wasn't published until 1931 (colour spaces will be described later). Apple incorporated basic colour management into its operating system quite early on and was instrumental in establishing the International Color Consortium, or ICC (**www.color.org**) in 1993. Apple's early support of colour control helped to establish its products in the field of art and design, and area in which the company has had considerable success. Microsoft Windows did not provide any colour management facility until Windows 98, and colour management tools have only recently appeared in Linux.

One of the problems in understanding colour management is that there are numerous standards and methodologies, each a product of the time it was introduced and the particular colour problems it was intended to address. For example, instead of just one colour model or space there are several, which can be confusing to the colour management novice. Another problem is that colour and colour management is a complicated subject and a great deal of the available information is either misleading or just plain wrong.